Fundamentals of Biomedical Engineering

(21BMO121T)

Open Elective subject for ECE

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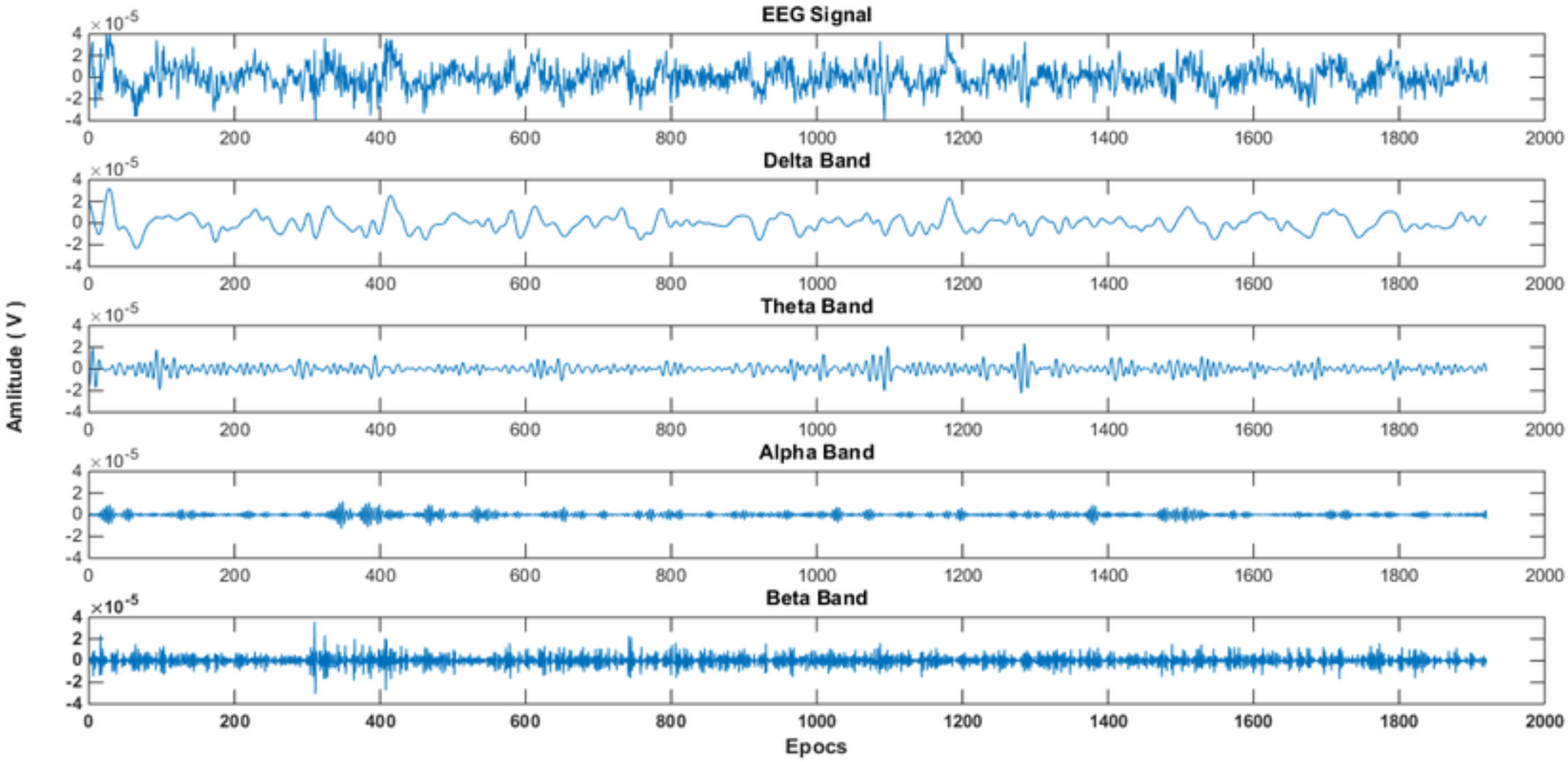
UNIT-1

Bioelectric signals – Electro-Encephalo-Gram EEG

Session-6_SLO-1

Bioelectric signals – Electroencephalogram



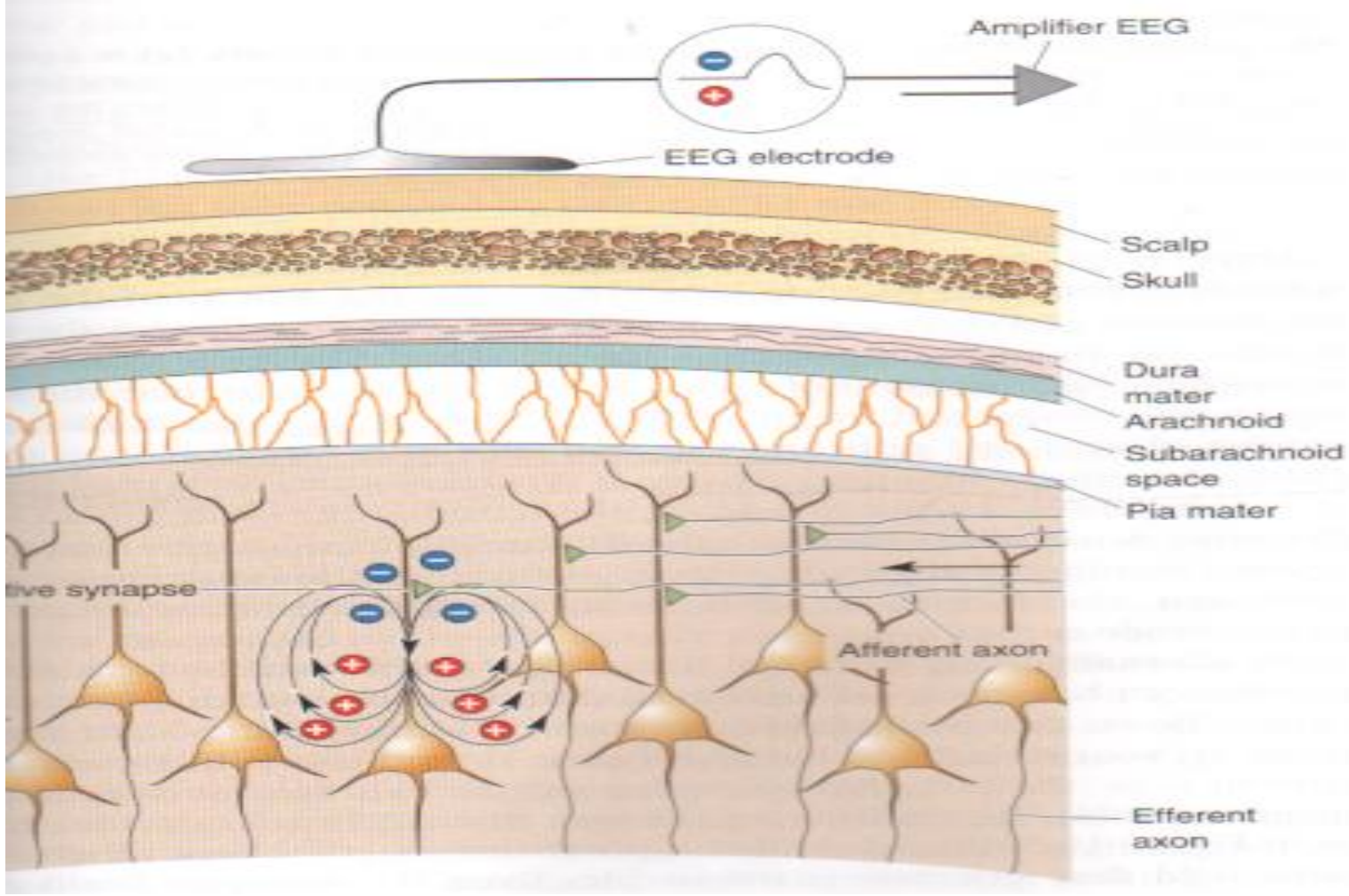
Bioelectric signals – Electroencephalogram

- The brain generates **rhythmical potentials** in the **individual neurons** of the brain.
- These potentials get **summated** and appear as a **surface waveform**, the recording of which is known as the **ELECTROENCEPHALOGRAM**.
- The neurons are **electrically polarized** at rest.
- The interior of the neuron is at a potential of about **–70 mV** relative to the exterior.
- When a neuron is **exposed to a stimulus** above a certain threshold, a nerve impulse, is generated which spreads in the cell resulting in the **depolarization** of the cell.
- Afterwards, **repolarization** occurs.

Bioelectric signals – Electroencephalogram

- The **EEG signal** can be **picked up** with **electrodes** either from the **scalp** or directly from the **cerebral cortex**.
- The peak-to-peak amplitude of the waves from the **scalp** is normally **100 μV** or less while that on the **exposed brain**, is about **1 mV**.
- The **normal EEG frequency** content ranges from **0.5 to 50 Hz**.
- The nature of the **wave varies** over the **different parts** of the scalp.

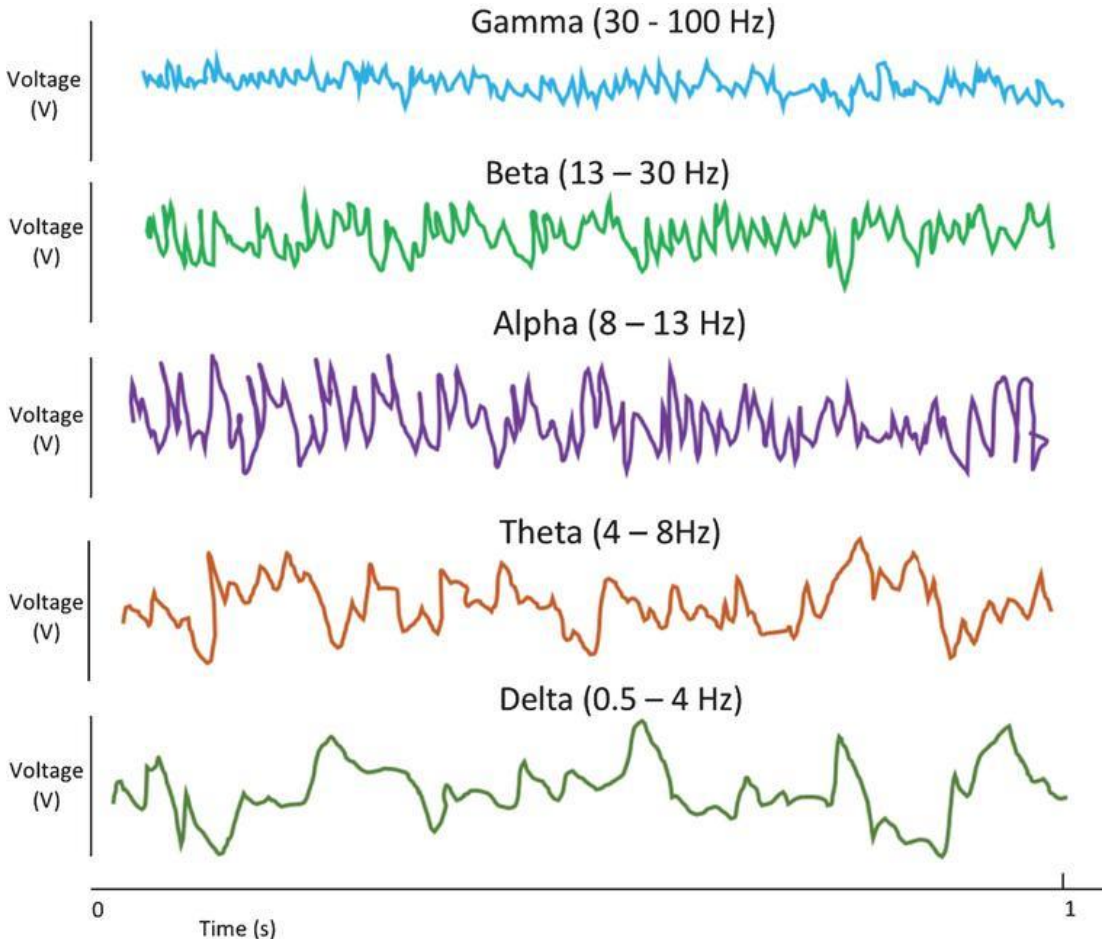
Origin of EEG waves



EEG is the record of electrical activity of brain (**superficial layer i.e. the dendrites of pyramidal cells**) by placing the electrodes on the scalp.

Bioelectric signals – Electroencephalogram

- The basic frequency of the EEG range is classified into the following five bands for purposes of EEG analysis:



Delta (δ)	0.5–4 Hz
Theta (θ)	4–8 Hz
Alpha (α)	8–13 Hz
Beta (β)	13–22 Hz
Gamma (γ)	22–30 Hz

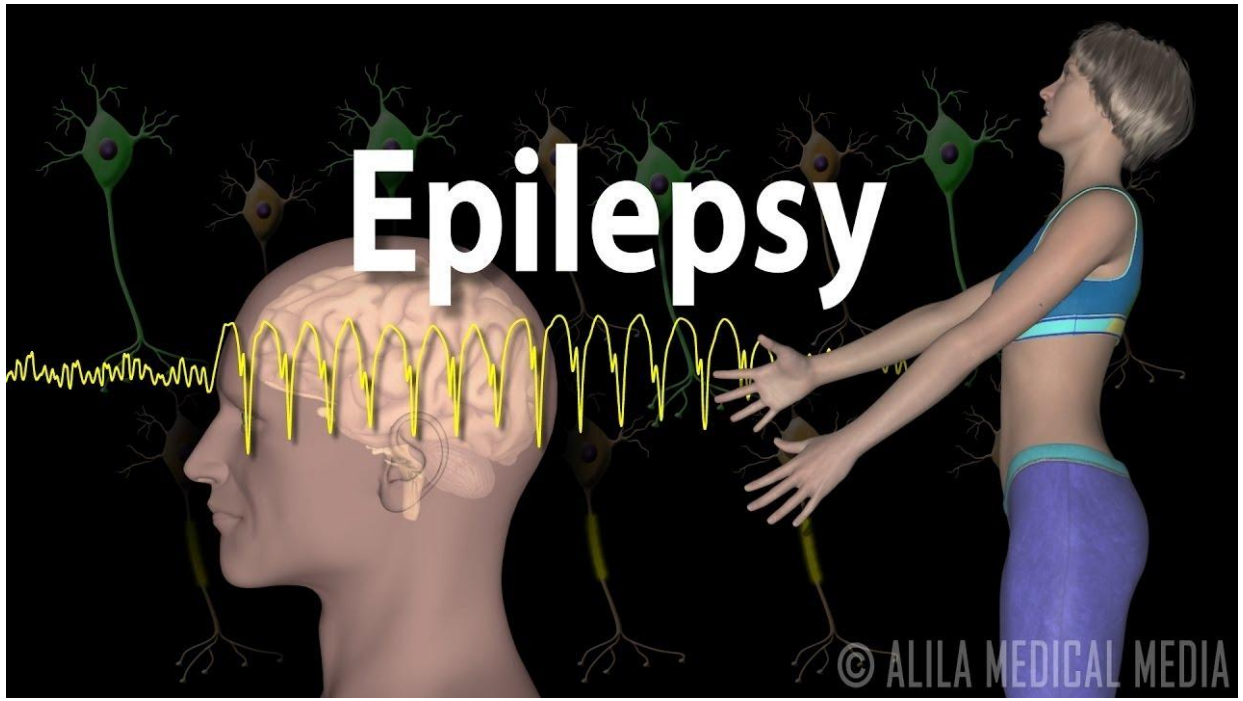
https://www.youtube.com/watch?v=h5LT9b-L9_Y

ALPHA WAVE -α

- rhythmic, **8-13 Hz**
- mostly on **occipital lobe**
- One of the principal components to indicate the state of '**alertness**' of the brain.
- It serves as an indicator of the **depth of anesthesia** in the operating room.
- The frequency affected by the mental activity of a person.

ALPHA WAVE -α

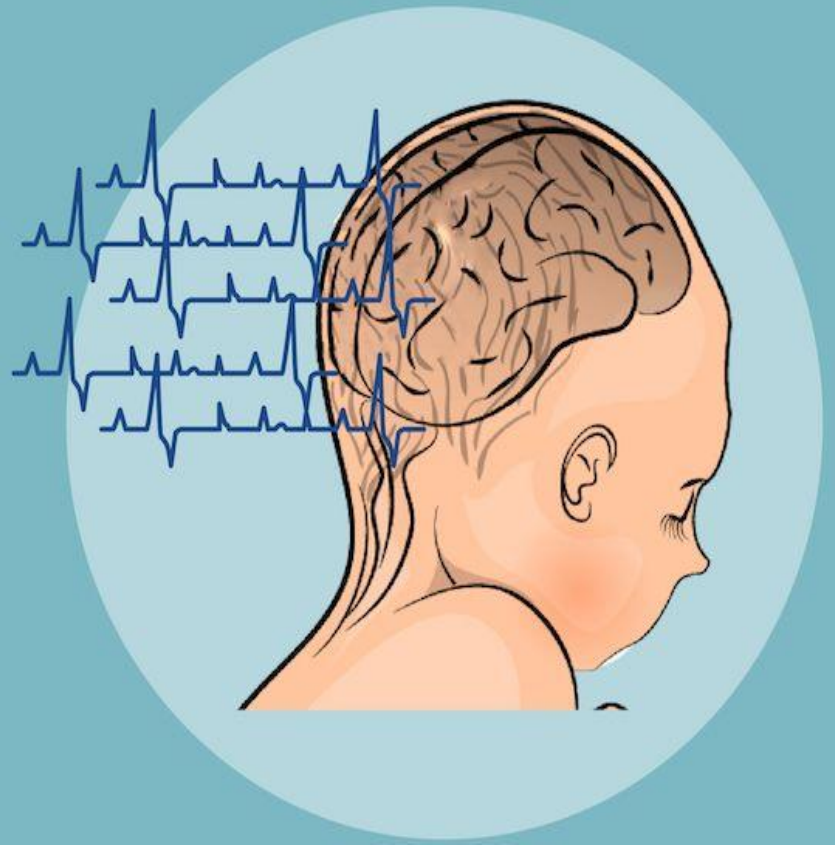
- **Lack of repeatability** in a given person from one occasion to another makes the analysis a difficult proposition.
- Abnormalities like **epileptic seizures** and **sleep disorders**.



WHAT ARE SEIZURES?

Seizures occur when there is an abnormal surge of electrical activity in the brain due to chemical changes in nerve cells.

Typically, there is a balance between **excitatory** and **inhibitory** brain cells. Excitatory brain cells cause activity, and inhibatory brain cells prevent it. Seizures can occur when this balance is disrupted and there is either too much or too little activity.



Evoked Response

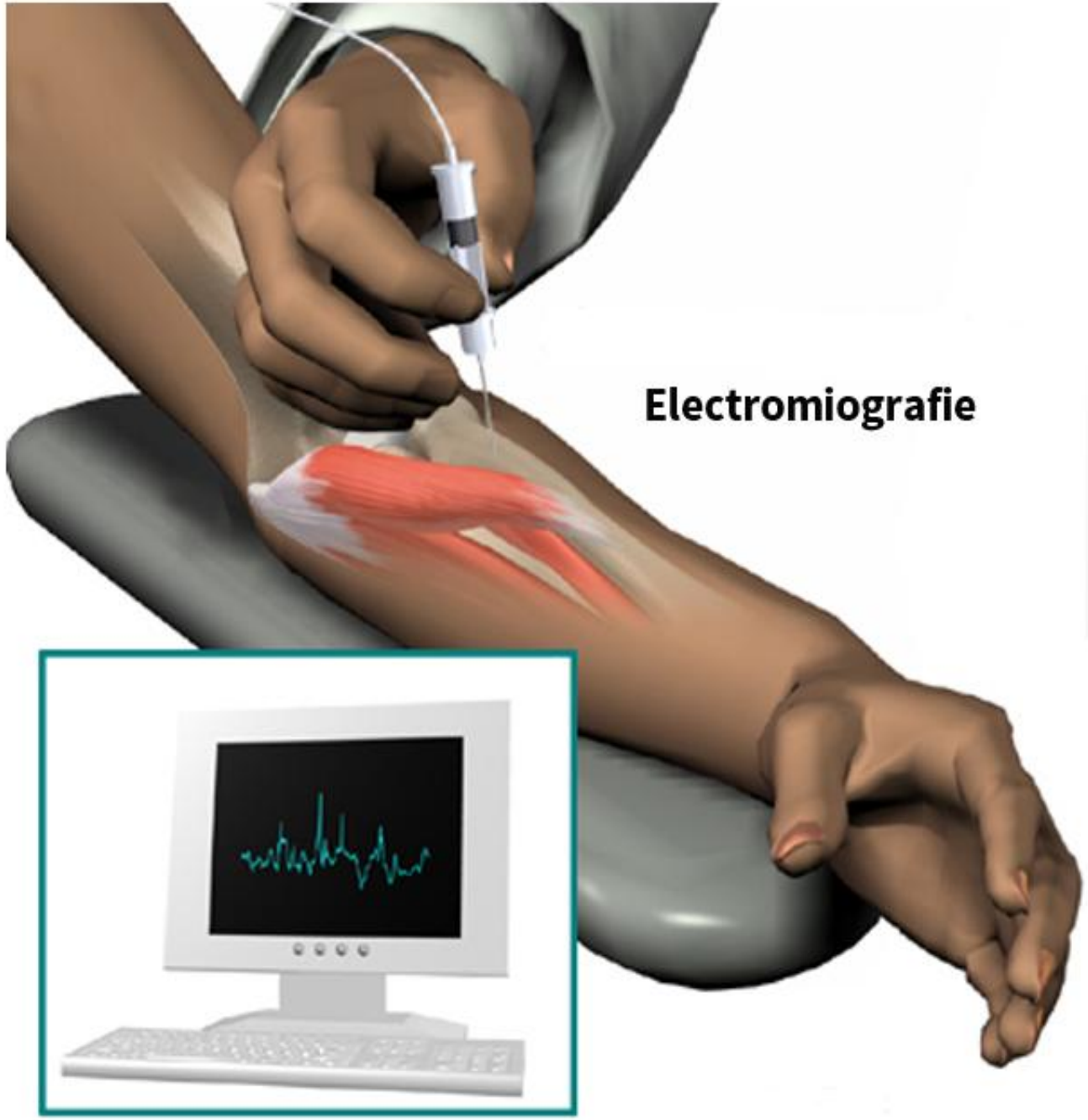
- Recording of '**evoked response**', which indicates the disturbance in the EEG pattern resulting from **external stimuli**. The stimuli could be a **flash of light** or a click of **sound**.
- The evoked response can be distinguished from the rest of the EEG activity.

UNIT-1

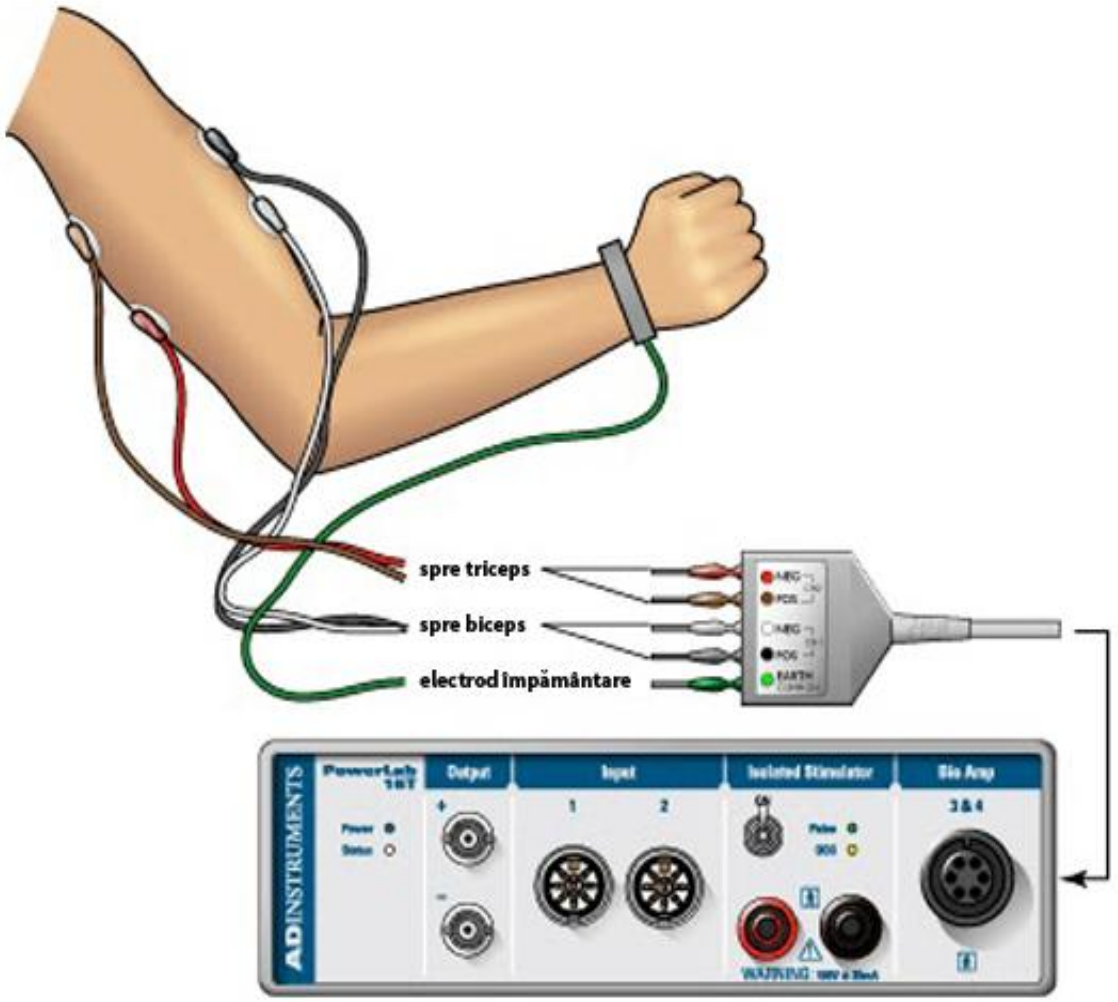
Bioelectric signals –EMG

Electro-Myo-Gram

Session-6_SLO-2



Electromiografie



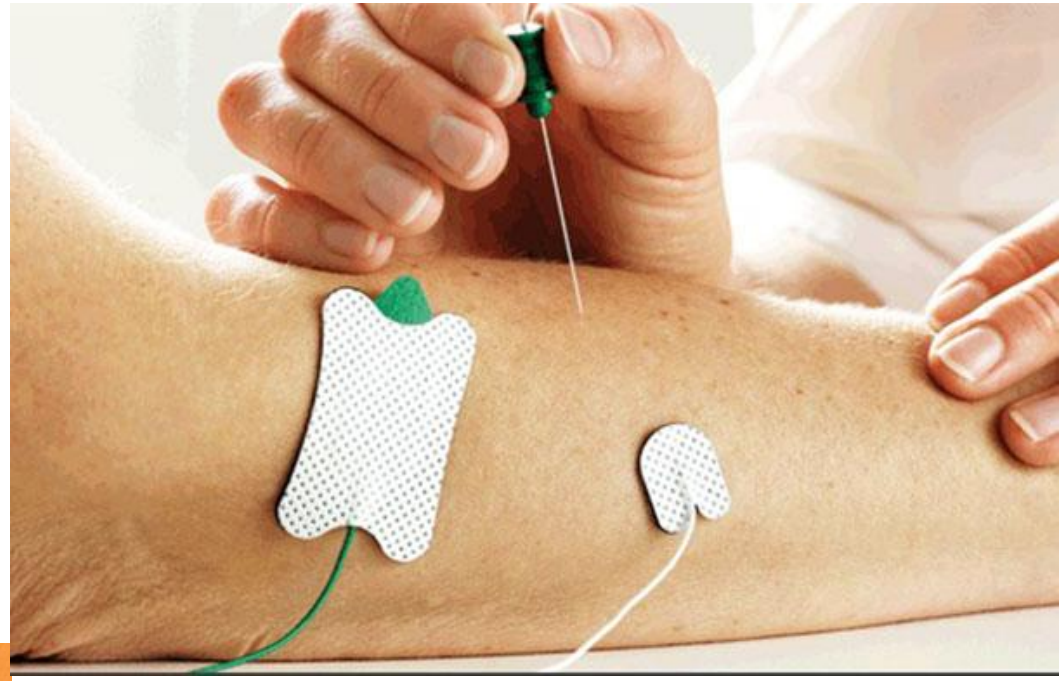
Bioelectric signals – Electromyogram

- The **contraction** of the **skeletal muscle** results in the generation of **action potentials** in the individual muscle fibers.
- The **activity** is similar to that **observed** in the **cardiac muscle**, and **skeletal muscle**.
- In the **skeletal muscle, repolarization** takes place much **more rapidly**, the action potential lasting only a few **milliseconds**.
- EMG pattern is usually a **summation of the individual action potentials** from the fibers constituting the muscle.
- The **electrical activity** can be **observed** by means of **surface electrodes** on the skin.

Bioelectric signals – Electromyogram

- Using **needle electrodes**, the action potentials from **individual motor units** can be recorded for **better diagnostic** information.
- **Voluntary muscle the potentials** range from **50 µV** to **5 mV** and the duration from **2 to 15 ms**.
- The **values vary** with the anatomic **position of the muscle** and the **size** and **location** of the electrode.
- In a **relaxed muscle**, there are normally **no action potentials**.

Bioelectric signals – EMG electrodes



Needle electrode

Surface electrode